

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: CSA04 **COURSE CODE:** Operating Systems

COURSE OUTCOME COVERED

CO1: Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks: 25

Annexure A

Technical Presentation

Code of Conduct:

I Hema Priya-T (192524467) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Hemapriya

Annexure A

Technical Presentation

Question 1 (5 Marks)

Suppose an Operating System is used in a banking system handling thousands of transactions simultaneously.

Prepare a technical presentation explaining:

Types of OS suitable for this scenario

OS structure used

Role of system calls in transaction processing

Question 2 (5 Marks)

Consider a multi-user Linux-based server environment supporting file sharing and device access.

Prepare a technical presentation covering:

OS functions involved in resource management

File and device system calls

Interaction between user processes and kernel

Question 3 (5 Marks)

Assume a real-time embedded system used in traffic signal control.

Prepare a technical presentation explaining:

Type of OS used and justification

OS structure and components

System calls used for process control and timing

Question 4 (5 Marks)

A company uses a distributed computing system across multiple locations.

Prepare a technical presentation discussing:

Type of OS and its characteristics

Architecture of distributed OS

System calls used for communication and coordination

Question 5 (5 Marks)

Consider an OS managing process scheduling, memory allocation, and I/O operations in a modern computer.

Prepare a technical presentation including:

Key OS functions and their importance

Interaction between OS components

Examples of system calls for each function

RUBRICS FOR CALCULATION

Technical Presentation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Question No.	Technical Content Accuracy (30%)	Problem Identification (25%)	Use of Tools (20%)	Communication (15%)	Professionalism (10%)	Total Marks
Q1						
Q2						
Q3						
Q4						
Q5						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Operating Systems in Action

Technical Presentation – CSA04 Operating Systems (CO1)

Assessment Tool 3

HEMA PRIYA T
192524467

Q1: Banking Transaction System

OS Type Suitable

- Multiprogramming / Time-sharing OS for concurrent transactions
- Multiprocessor OS to handle heavy load across CPUs
- Network OS connecting branch and central servers

OS Structure Used

- Client-Server architecture (ATM/App as client, bank server as host)
- Layered / Microkernel structure for security and reliability

Role of System Calls

- `fork()/exec()` - create process for each transaction
- `read()/write()` - update transaction records
- Locking calls - maintain data consistency



Q2: Multi-User Linux Server

OS Functions in Resource Management

- Process management - handles multiple user sessions
- File management - permissions and ownership per user
- Device management - controls access to shared devices



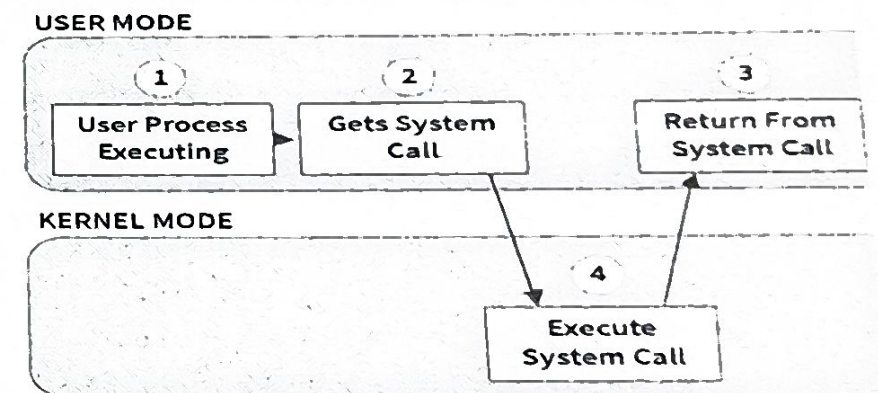
File and Device System Calls

- `open()`, `read()`, `write()`, `close()` - file operations
- `chmod()`, `chown()` - permissions and ownership
- `ioctl()` - device control commands

Kernel Interaction

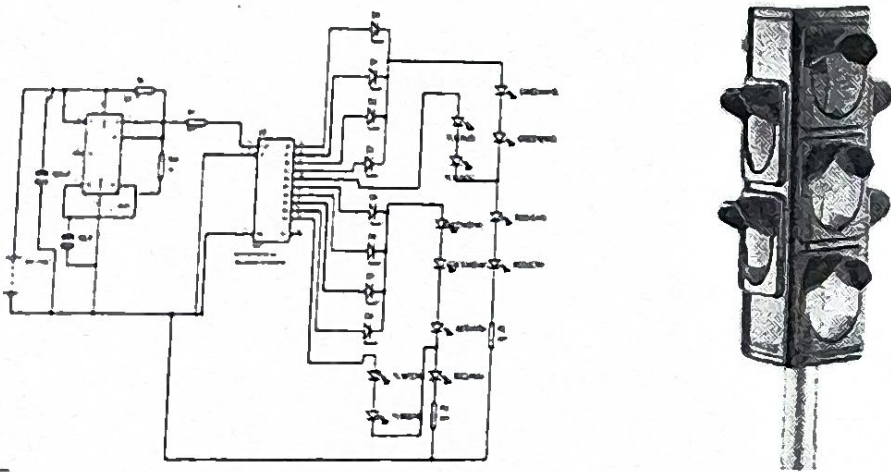
- User process makes system call -> switches to kernel mode
- Kernel executes request and returns control to process

WORKING OF A SYSTEM CALL



Q3: Real-Time Traffic Signal Control

Four Way Traffic Lights



Type of OS and Justification

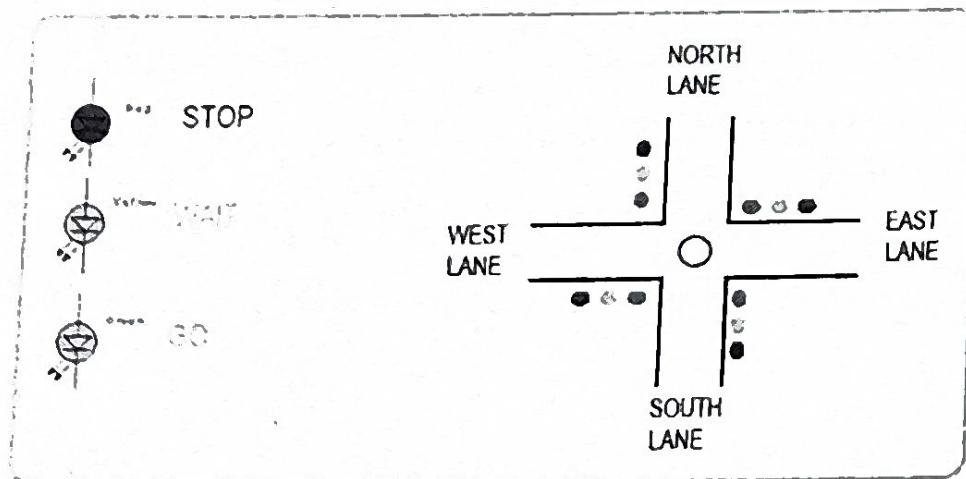
- Real-Time Operating System (RTOS)
- Needed for predictable, time-bound responses

OS Structure and Components

- Microkernel structure - small and fast
- Components: scheduler, interrupt handler, timer manager

System Calls Used

- `task_create()` - create timing-critical task
- `sleep()/delay()` - hold signal state for fixed duration
- `set_priority()` - assign task priority



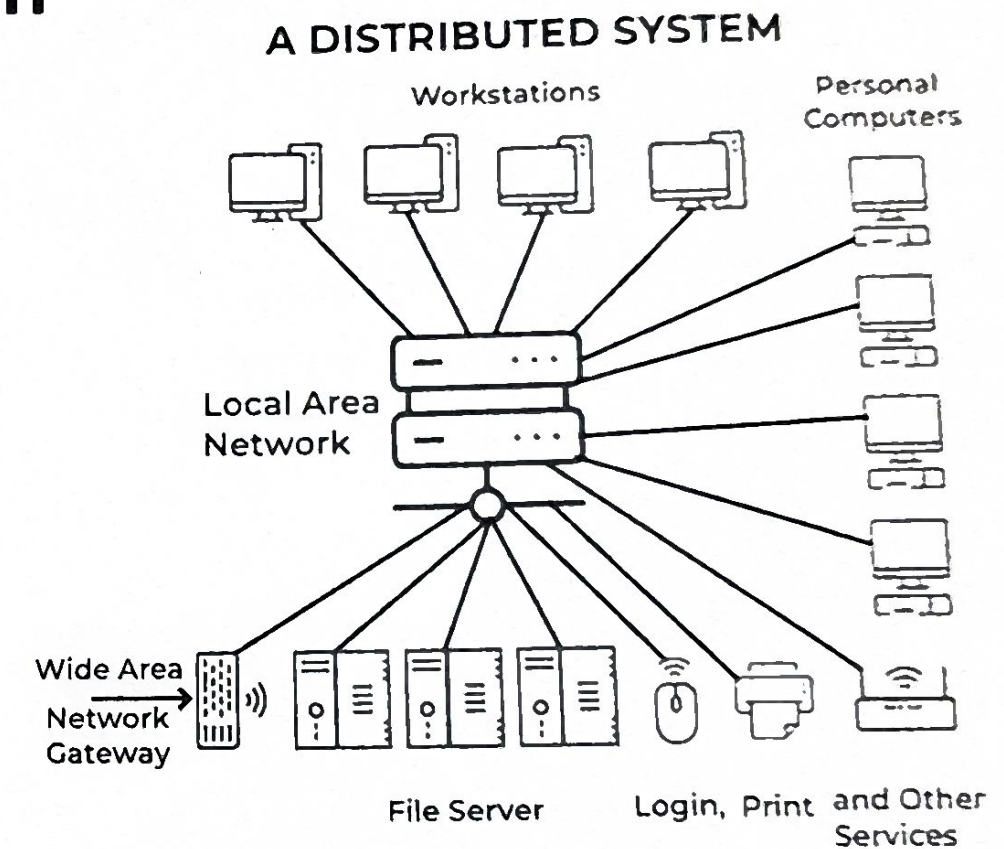
Q4: Distributed Computing System

Type of OS and Characteristics

- Distributed Operating System (DOS)
- Features: transparency, scalability, fault tolerance

Architecture of Distributed OS

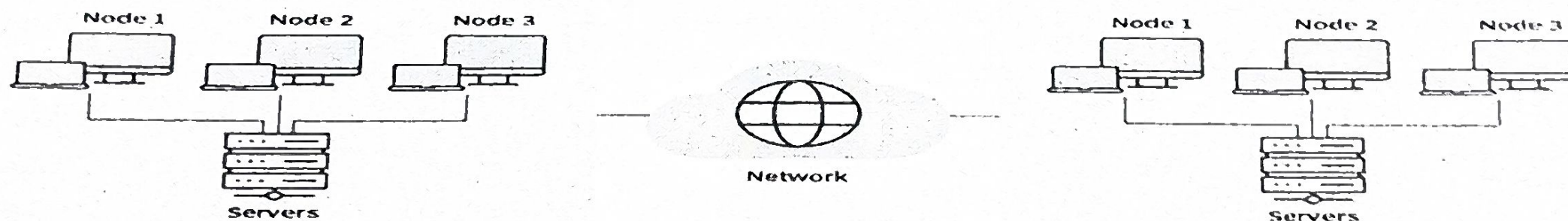
- Client-Server or Peer-to-Peer model
- Middleware layer connects multiple locations



System Calls for Communication

- `socket()`, `send()`, `recv()` - network communication
- RPC (Remote Procedure Call) - execute remote tasks

The distributed computing process



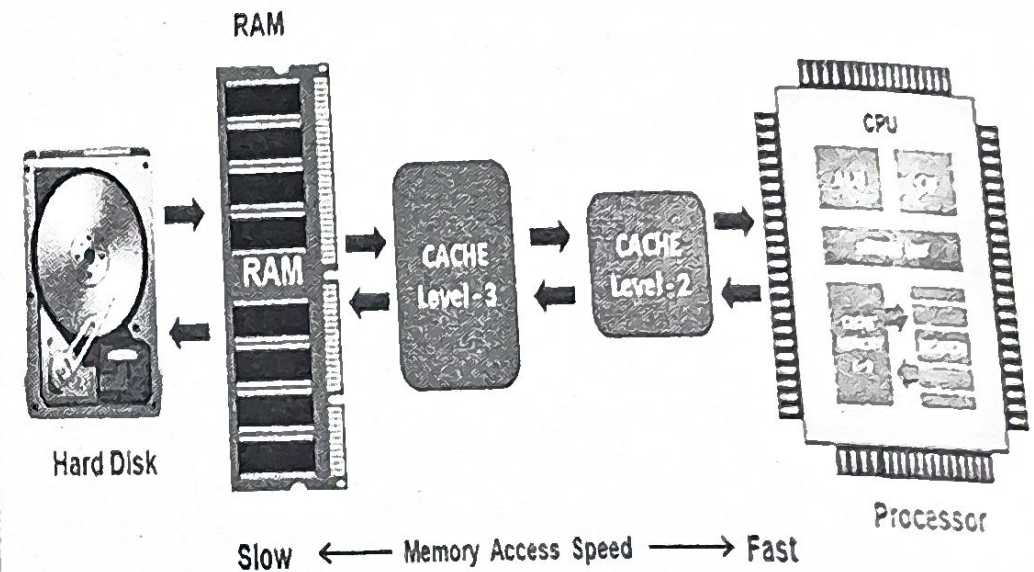
Q5: General-Purpose Modern OS

Key OS Functions

- Process scheduling - manages CPU time
- Memory allocation - manages RAM usage
- I/O operations - manages input/output devices

Interaction Between Components

- Scheduler, memory manager, and I/O manager work together
- System calls connect applications to these functions



Example System Calls

- `fork()`, `wait()` - process control
- `brk()`, `mmap()` - memory management
- `read()`, `write()` - input/output

